

Flux: The Elegant Machine Learning Library Written in Julia



The machine learning landscape is filled with a large number of libraries, each of which comes with its own pros and cons. Choosing the right tool for your application is an important decision. This article explores Flux – a Julia library that provides you the right mix of abstraction and fine-tuning. Flux is fully written in Julia and comes with prominent features such as excellent GPU support and differentiable programming.

Machine learning has evolved into a prominent paradigm of developing solutions to problems. It has enabled solving problems that were earlier unsolvable with relative ease. In the machine learning domain there are plenty of tools and libraries to choose from. Selection of the right tool contributes to the success of the application in a significant manner.

Each of the libraries associated with machine learning has its own advantages and disadvantages. Some of the libraries allow you to work at only the given level of abstraction. If you want to perform some sort of tweaking under the hood to optimise performance, these libraries make it either very difficult or near impossible.

In this backdrop, Flux is an elegant machine learning stack which is fully written in Julia. If you are absolutely new to Julia, feel free to explore the earlier articles on Julia in this magazine. In a nutshell, Julia is a language that walks like Python and runs like C. It has the advantages of both Python (by making the implementation simpler) and of C (by making the code run faster).

Being fully written in Julia, Flux is guaranteed to run at comparatively greater speed than its counterparts. Another big advantage is that it allows you to work at a great level

of abstraction if you wish to do so. At the same time, if you want to go under the hood and tweak the algorithm, it enables you to do so as well. In the words of the official documentation, Flux makes the easy thing easy while remaining fully hackable (<https://fluxml.ai/>).

The Flux features

Flux comes loaded with features, which as a machine learning engineer you would always prefer to have (Figure 1).

- Optimised for performance.
- Ability to perform differentiable programming. It has state-of-art models such as Neural ODE and Zygote. Automatic differentiation (AD) in Flux can be achieved through its advanced system called *Zygote*. Computing gradients can be done effortlessly and optimally with Flux.
- Good support for GPU, which is accomplished through *CUDA.jl*. It comes with fully tweakable support.
- Model Zoo provides a great collection of Flux scripts, which can be utilised to customise as per your needs. Pretrained Flux models are also available with TextAnalysis and Metalhead.
- Support for TPU is an important feature of Flux.